Thematic Analysis of Classic Adventure Novels Using Natural Language Processing Techniques

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Abstract

The aim of this study is to use natural language processing (NLP) techniques to explore thematic patterns in classic adventure novels. This report applies NLP methods to analyse six well-known works from the genre. Techniques such as word frequency, word clouds, TF-IDF, topic modelling, and sentiment analysis are used to examine language patterns, emotional tone, and thematic structures. The findings reveal common themes such as exploration, danger, and survival, along with distinct narrative elements unique to each novel. These results contribute to a deeper understanding of how the adventure genre has evolved over time.

1 Introduction

Adventure novels have been a popular genre since the early 1700s, telling stories of exploration, survival, and heroism. Comparing adventure novels from different periods may provide valuable insight into how language, writing styles, and storytelling have changed over time. This project focuses on adventure novels from the past. Adventure novels are considered, rather than any other genre, because of their distinctive narrative structure, thematic richness, and historical significance in shaping literary traditions. This gives us the ability to analyse various patterns and themes in novels across decades. The six selected novels were: *Robinson Crusoe* (1719), *Moby-Dick* (1851), *Treasure Island* (1883), *King Solomon's Mines* (1885), *The Call of the Wild* (1903), and *The Lost World* (1912). These works represent a broad span of time, each reflecting a different theme.

Traditional literary analysis is limited by its reliance on human interpretation, which can be subjective and time consuming. Natural language processing (NLP) can process large volumes of text and reveal trends and patterns that may not be apparent through closed reading. Hence, NLP techniques allow for a more detailed analysis of text documents and offer valuable insights.

This study uses NLP techniques, including word clouds, TF-IDF, topic modelling and sentiment analysis. These methods help uncover changes in word usage, emotional tone, and narrative patterns over time. This research will explore how language and themes in adventure fiction evolved from the 17th century to the early 20th century. This approach provides a detailed overview of how classic adventure novels reflect the eras they were written in and how they have shaped modern literature. The code for this project can be found in the following Github repository: URL: Click here to visit: https://github.com/notprat543/MDS-FINAL-REPORT.git

1.1 Change in approach

In the previous trimester, the progress report presented a more individualised analysis, focusing on each book separately. This trimester, the analysis is conducted on the entire dataset as a whole. As in the previous trimester, individual sentiment progression graphs for each novel were produced, however, this time they were scaled on the same graph and analysed using SenticNet. Additionally, protagonist mentions and NRC emotion categories were introduced into the analysis.

2 Background

NLP is a widely used tool in literary studies. It offers methods to extract and analyse patterns in text that would be difficult to identify through traditional close reading. In the last decade, NLP has been used to analyse all kinds of documents, including poems, novels, and historical texts.

One of the most prominent applications has been in uncovering thematic structures and supporting recommendation systems for literature (Lundy 2020; Devika &Milton 2025). NLP techniques have been applied in a variety of literary studies. For example, Crutchfield used them to categorize fictional literature, while Zhang and Gao used them to compare stylistic characteristics between poets (Crutchfield 2020; Zhang & Gao 2017). Beyond textual analysis, it has also been used for profiling fictional characters (Jacobs 2019).

Although there have been efforts to compare novels, many studies have relied on qualitative analysis or basic textual metrics rather than NLP techniques (Hunt 2001). This limits the depth of the analysis and makes it difficult to identify subtle themes and narrative structures. There are examples of computational studies that include novels. However, most of them focus on a single work or use them as examples within larger datasets. (Yuan, Vengadasamy & Zheng 2025; Hoque et. al. 2023). A noticeable gap in the literature is the absence of genre-specific, comparative analyses that span several decades and apply modern NLP tools.

This study aims to address that gap by focusing specifically on the adventure genre. Adventure novels have been a popular and influential genre for centuries, known for themes of exploration, survival, danger, and discovery. These stories often reflect the values, fears, and ambitions of the time in which they were written. Analysing them with NLP allows us to see which parts of the genre have stayed the same and which have changed. This study focuses on six well-known adventure novels from 1719 to 1912, using word clouds, TF-IDF, topic modelling, and sentiment analysis to compare their language and themes. Through this approach, the study offers valuable insights into the evolution of adventure genre over time and highlights how various authors have contributed to its development.

3 Methods

3.1 Data Cleaning and preparation

The data used for this project was sourced from Project Gutenberg, which provides access to literature in the public domain. Each novel was downloaded into R as plain text files and transformed into a data frame. Each text included additional material, such as Project Gutenberg's headers, licensing information, and footers, which are not part of the novel content and could interfere with analysis if left unaddressed. These were removed from each novel by identifying the appropriate line indices. Subsequently, the texts were organised into structured data frames consisting of six columns:

Table 1: Dataset

Gutenberg ID	Project Gutenberg catalogue number
Text	The main body of the text
Book	The title of the novel to which the line belongs
Line number	Line number in the original order of the text
Chapter	Chapter identifier
Part	Part identifier

This metadata will be useful in the analysis. Furthermore, all empty lines were removed from the dataset to facilitate tokenisation and frequency analysis. To focus on the meaningful language and reduce noise, common words such as "the" , "and" and many others were removed from the dataset. Although these words are important to sentence structure and grammar, they add no value to the thematic analysis and were removed to improve clarity. These words were removed from the dataset using the stopwords package in R. Chapter and part numbers were also removed from the texts. All novels were merged into one dataset for analysis.

Furthermore, a custom list of words (see Appendix A) was removed from the dataset. This list includes proper nouns like character names, locations, and other frequently repeated words. These words were removed as they carry no value to the thematic analysis and dominate frequency charts. Additionally, all punctuation was removed from the dataset for consistency.

In word frequency visualisations, words like "run" and "running" may appear separately, even though they represent the same underlying theme. To address this, lemmatisation was performed to reduce words to their base or dictionary forms (e.g., "running" to "run" and "dogs" to "dog"). Furthermore, a custom list of words was replaced with their semantic forms (see Appendix A).

This process helped consolidate similar words and improved clarity for both topic modelling and frequency analysis.

3.2 Word clouds

After the initial data cleaning, the next stage involved exploring word frequencies and their visualisations. Word frequencies provide insight into how often each word is used. Following data cleaning and preparation, the frequency of each unique word within each novel was calculated using the `count` function.

Word clouds were chosen as the primary method of visualisation. They provide a simple and clear representation of the most frequent words. In this format, the most common words appear at the centre of the graph and are displayed in a larger size. This approach enables us to clearly identify prominent themes in each book. These results are displayed in section 4.1.

3.3 TF-IDF

We used Term Frequency-Inverse Document Frequency (TF-IDF) to identify words that were significant in individual books i.e., the most distinct words in each book. It measures the importance of a word within a particular document relative to its occurrence across all documents. The classical formula for the TF-IDF score of the i^{th} term in the j^{th} document is given by:

$$\text{TF-IDF}(i,j) = \text{tf}(i,j) \times \log\left(\frac{N}{\text{df}_i}\right) \tag{1}$$

where N represents the total number of documents in the collection. The term frequency $tf(i,j)$ denotes how often term i appears in document j, while df_i is the number of documents in which term i appears (Dennis-Henderson 2020, p. 71) . Words that are either rare within a document or common across all documents are assigned low TF-IDF scores. This approach helps distinguish each novel by highlighting distinct themes and patterns for every novel.

TF-IDF scores were calculated for every word in each of the six novels, and we focused on extracting the most significant terms. The top 10 words with the highest TF-IDF scores were selected for each book, and these terms were visualised using comparative bar charts to show the top 10 words for each book based on their TF-IDF scores (see section 4.2).

3.4 Topic Modeling

Latent Dirichlet Allocation (LDA) was used to uncover thematic structures within these books. LDA is a topic modelling technique designed to address challenges such as missing data and overfitting. It is a probabilistic model that assumes each document is a mixture of topics, and each word is associated with one of those topics (Dennis-Henderson 2020, p.89). This model helps identify themes in the text by assigning probabilities to topics and words.

After pre-processing, the frequency of each word across the books was calculated. Initially, LDA was applied to the combined dataset to uncover topics shared across all the books. Furthermore, a Document-Term Matrix (DTM) is constructed to represent the frequency of each word across the documents. The LDA model is then applied to the DTM using the `LDA()` function from the `topicmodels` package. Hence, for each topic, the top terms with the highest probabilities are visualised using a bar plot . Subsequently, the LDA model was applied separately to each book. This allowed for the identification of internal thematic variations within each novel. Each plot provides insight into the topics that are dominant within the books (see section 4.3).

3.5 Sentiment Analysis

Sentiment analysis is used to check the emotional theme of a document. It was a critical part of this study, aimed at examining the emotional tone present in each novel. In this analysis dictionary methods were used. Dictionary methods are unsupervised and very efficient for large texts due to their straightforward scoring systems and its effectiveness in handling the clear, traditional language styles . For this study, we compared the coverage of various dictionaries with our dataset i.e., the proportion of words that overlap between each dictionary and our dataset. Senticnet had the highest proportion of coverage and was selected for use. This dictionary assigns a numeric value between -1 to +1, with negative sentiments closer to -1 and positive sentiments closer to +1.

Once the data was pre-processed, an inner join was used to match words from the novels to those in the senticnet lexicon. Words without an associated sentiment score were excluded from further analysis. To enable better comparison across books, the chapters were scaled using the following formula

$$x_{\text{new}} = \left((\max_{\text{new}} - \min_{\text{new}}) \times \frac{x_{\text{old}} - \min_{\text{old}}}{\max_{\text{old}} - \min_{\text{old}}} \right) + \min_{\text{new}} \tag{2}$$

This transformation mapped the chapters to a continuous scale ranging from 1 to 10, making the results easier to interpret across different books (Dennis-Henderson 2020, p.104).

Two levels of sentiment analysis were conducted. First, the average sentiment score for each book was calculated and visualised using a bar chart, allowing for a comparative overview across the novels

Next, sentiment scores were calculated at the standardised chapter level. Each standardised chapter's score was computed as a weighted average, where word frequency determined the influence of each term on the overall sentiment. These scores were visualised using the `plot_sentiment_by_book()` function. This function filters the dataset to include only words with associated sentiment scores and calculates the average sentiment score per chapter. Additionally, to smooth short-term fluctuations, rolling means of the standardised chapter scores were visualised as a linegraph showing sentiment variation throughout the narrative. This analysis helped trace the emotional arc within each novel and offered insights into how sentiment evolved over the course of the story (see section 4.4).

3.6 NRC word emotions

To understand the emotional tone of each novel, we used the NRC Emotion Lexicon. This method looks at eight emotions: anger, fear, anticipation, trust, surprise, sadness, joy, and disgust. It also includes overall positive and negative sentiment. We calculated the proportion of each emotion for every novel to allow comparison. These results were visualized using a bar chart, where bars were grouped by book to enable comparison (see Section 4.5).

3.7 Protagonist mentions

Protagonist mentions were calculated to investigate the main character's importance to the story. This method reflects the fluctuations in focus on the protagonists while highlighting the sections where they are central to the narrative. For each novel, a list of protagonist's names was created, including common variations and aliases (e.g., “Jim Hawkins”, “Jim”, “Hawkins” for *Treasure Island*). This ensured that all relevant references to the main character were captured, regardless of how they were written in the text. Subsequently, the dataset was filtered to retain only those words that matched with this list of protagonists. The number of mentions were calculated for each standardised chapter. This step allowed books of varying lengths to be compared on the same scale. To ensure that chapters with no protagonist mentions were still included, missing chapter values were filled with zero counts. A three chapter rolling mean was then applied to the data to smooth out short-term fluctuations and highlight broader trends in protagonist mentions over time.

These results were visualised using a scatterplot with a smoothed line for each novel (see section 4.6). This plot displays how the frequency of protagonist mentions changes across the course of each novel. This approach highlights chapters where the protagonist is most present and captures the fluctuations in focus that are characteristic of adventure storytelling, where the central hero’s journey is interspersed with secondary plotlines, descriptive passages, or shifts in perspective. Additional plots were generated for each novel, displaying a single graph that plotted protagonist mentions against chapter.(see Appendix A.3)

4 Results

4.1 Word clouds

Word clouds were created for each novel to display the hundred most common words (see Section 3.2). This graph provides an insight into the central themes that shape the narrative.

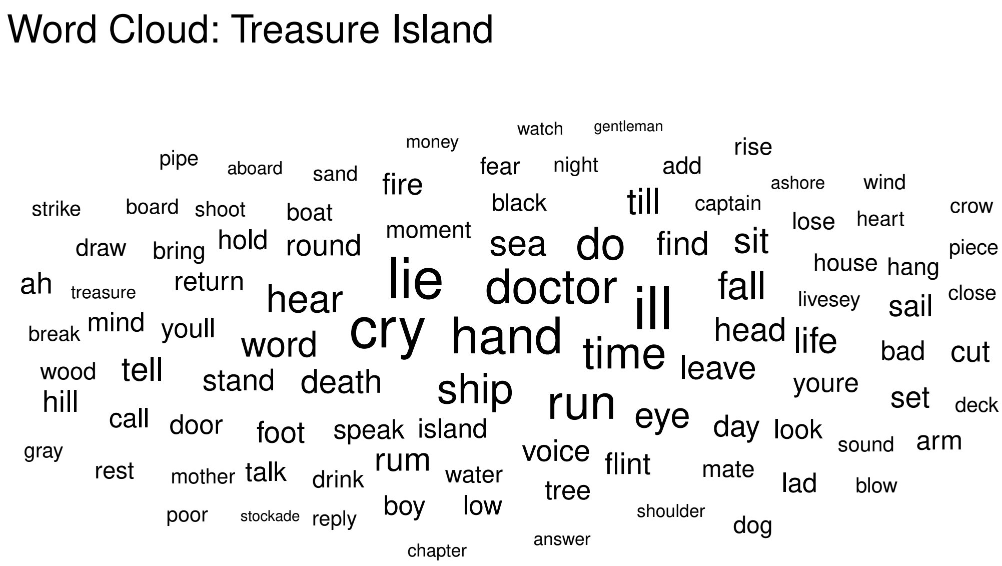

Figure 1: 100 most frequent words in *Treasure Island*

The most common words in *Treasure Island* (Figure 1) highlight the key themes of the novel, such as sailing and adventure. They indicate critical plot moments like betrayal or deceit, which are important to the story. Furthermore, there are many common words like "head" or "hand", which may indicate interactions between characters. This plot reinforces the themes of adventure and exploration, as well as displaying the emotional theme and interactions between characters.

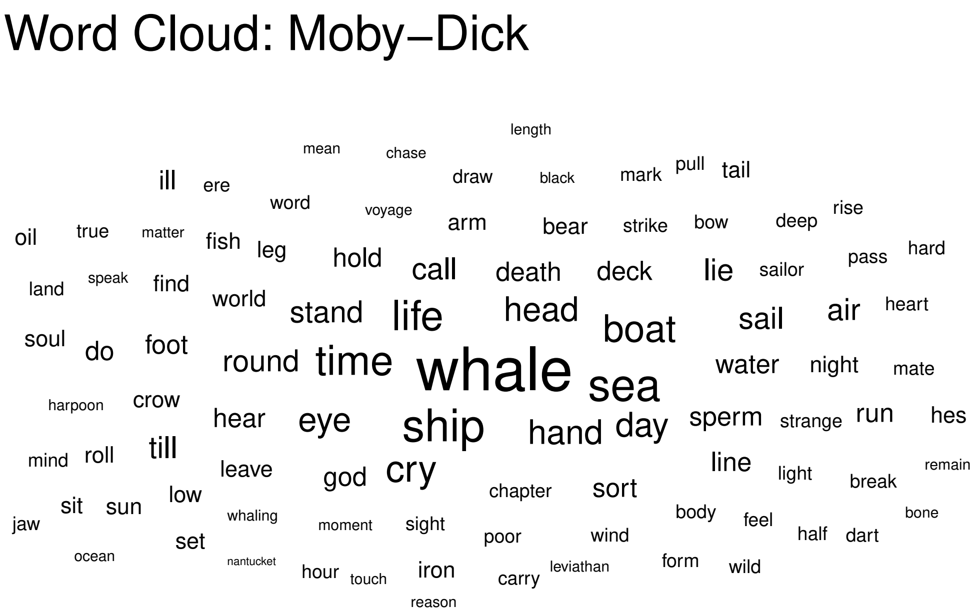

Figure 2: 100 most frequent words in *Moby-Dick*

(Figure 2) displays the most common words in *Moby-Dick*. The central themes of the sea, exploration, and the hunt for the whale are clearly evident in the graph. “Whale” is the most common word in the novel, emphasising its importance to the story. Furthermore, the graph includes several words that reinforce the theme of hunting the whale. This graph also highlights the emotional aspect of this adventure. These terms encapsulate the novel’s themes of obsession, hunting, and the characters’ journey through the ocean.

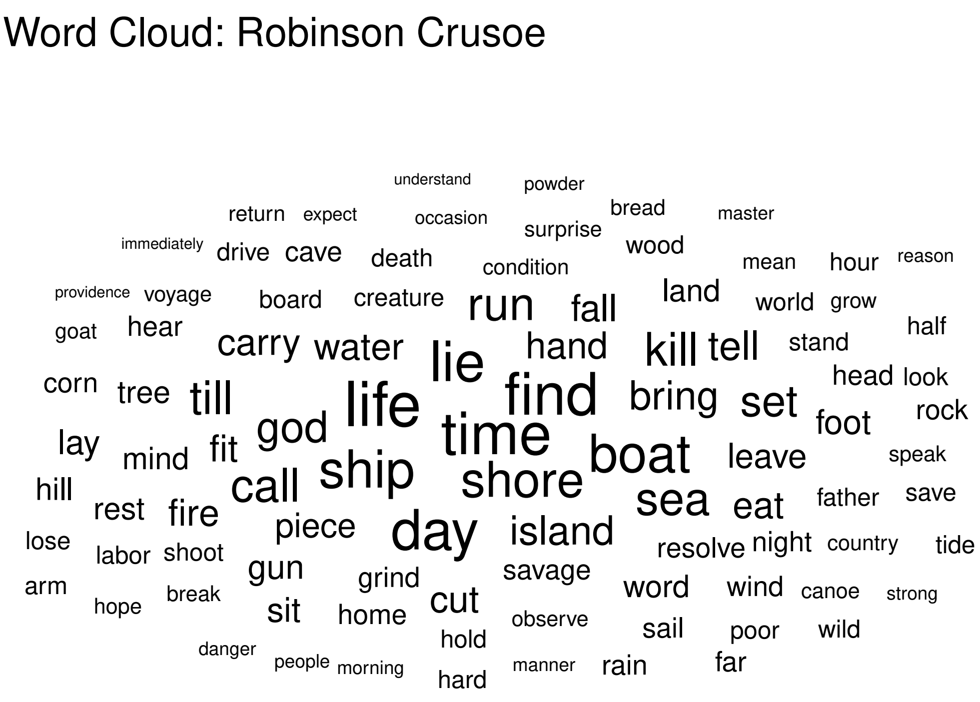

Figure 3: 100 most frequent words in *Robinson Crusoe*

The most frequent words in *Robinson Crusoe* are displayed in (Figure 3). Themes such as isolation and survival are evident in this graph. Furthermore, it emphasises the passage of time during the struggle for survival. The graph also highlights nautical elements and illustrates the themes of faith and perseverance.

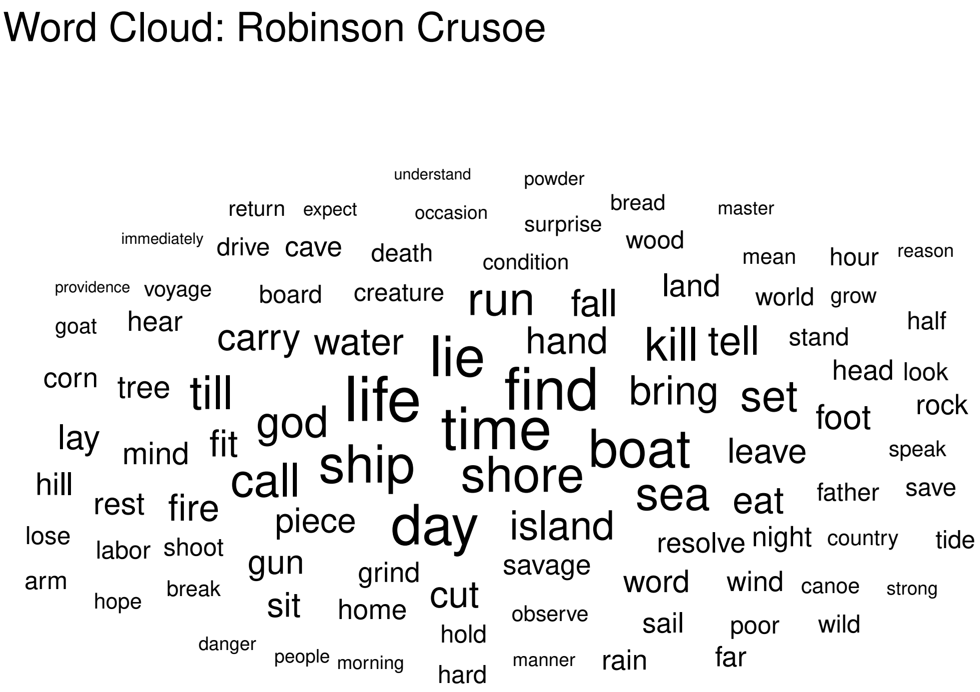

Figure 4: 100 most frequent words in *The Lost World*

Figure 4 highlights the prominent themes of discovery and exploration in *The Lost World*. It reflects the discovery of a new world filled with prehistoric animals and creatures. The graph also describes the setting of this world. It also emphasises interactions between characters and the emotional tone of the novel. Furthermore, it showcases the characters' sense of fear and caution throughout their journey of exploration and discovery.

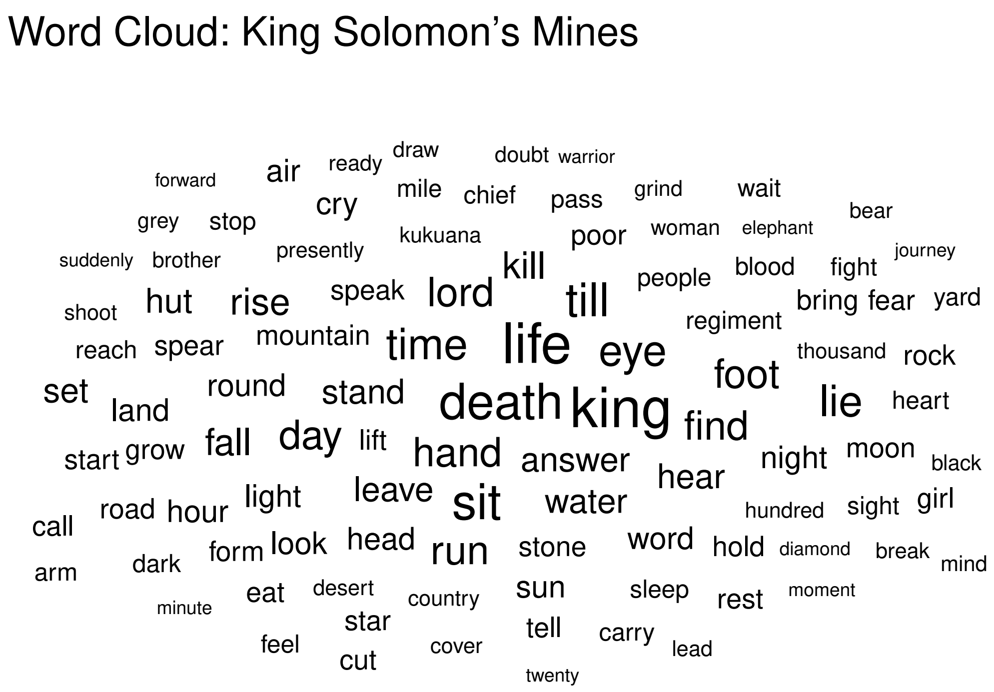

Figure 5: 100 most frequent words in *King Solomon's Mines*

As seen in Figure 5, *King Solomon's Mines* presents a narrative of a high-stakes adventure, highlighting dramatic challenges that often mean life or death for the characters. Terms like "regiment", "rock", "mountain", "shoot" and "hut" describe the harsh and rugged setting of the story. The graph also highlights the importance of the mines as a key location in the narrative. Furthermore, it emphasises the recurring themes of violence and conflict throughout the narrative. In addition, it reflects the constant danger and mystery of their journey.

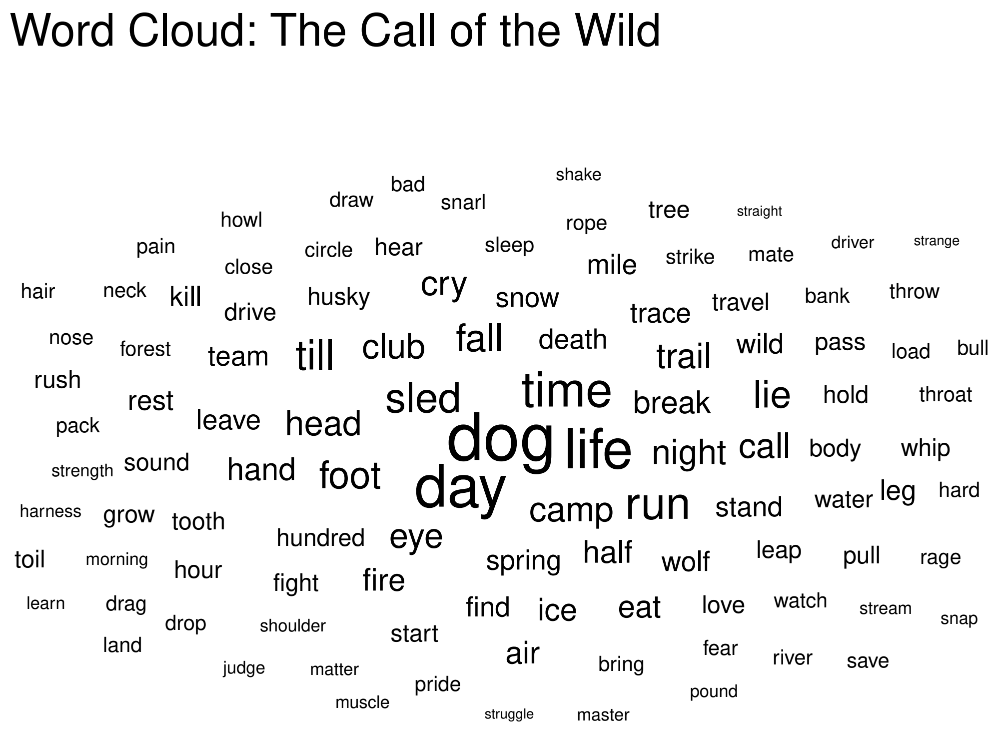

Figure 6: 100 most frequent words in *The Call of the Wild*

The most common words in *The Call of the Wild* are displayed in Figure 6. This graph highlights the central theme i.e., the protagonist's journey through the wild. It reflects the themes of survival and exploration while also conveying the emotional tone of the narrative. Words such as "rage", "struggle", and "cry" capture the intense emotional journey of the main character. Furthermore, the graph illustrates the forest landscape and emphasises the challenges of surviving in the wilderness.

Word cloud analysis of these six novels reveals shared and distinct themes across the genre. Exploration and adventure are a shared theme, but each novel explores the premise in a unique way. Additionally, *Treasure Island*, *Moby-Dick*, and *Robinson Crusoe* share a common theme of adventure at sea. *Treasure Island* emphasises themes of sailing. *Moby-Dick* showcases the relentless hunt for the whale, whereas survival and isolation are at the forefront of the narrative in *Robinson Crusoe* and *The Call of the Wild*. In contrast, *King Solomon's Mines* presents a rugged theme of a high-stakes expedition. Each novel reflects a recurring emphasis on its own interpretation of adventure, survival, and human endurance. The prominence of these themes are the defining elements of adventure fiction.

4.2 TF-IDF

TF-IDF scores help identify subtle themes in text. The top 10 TF-IDF scores were calculated for each of the six novels (see Section 3.3).

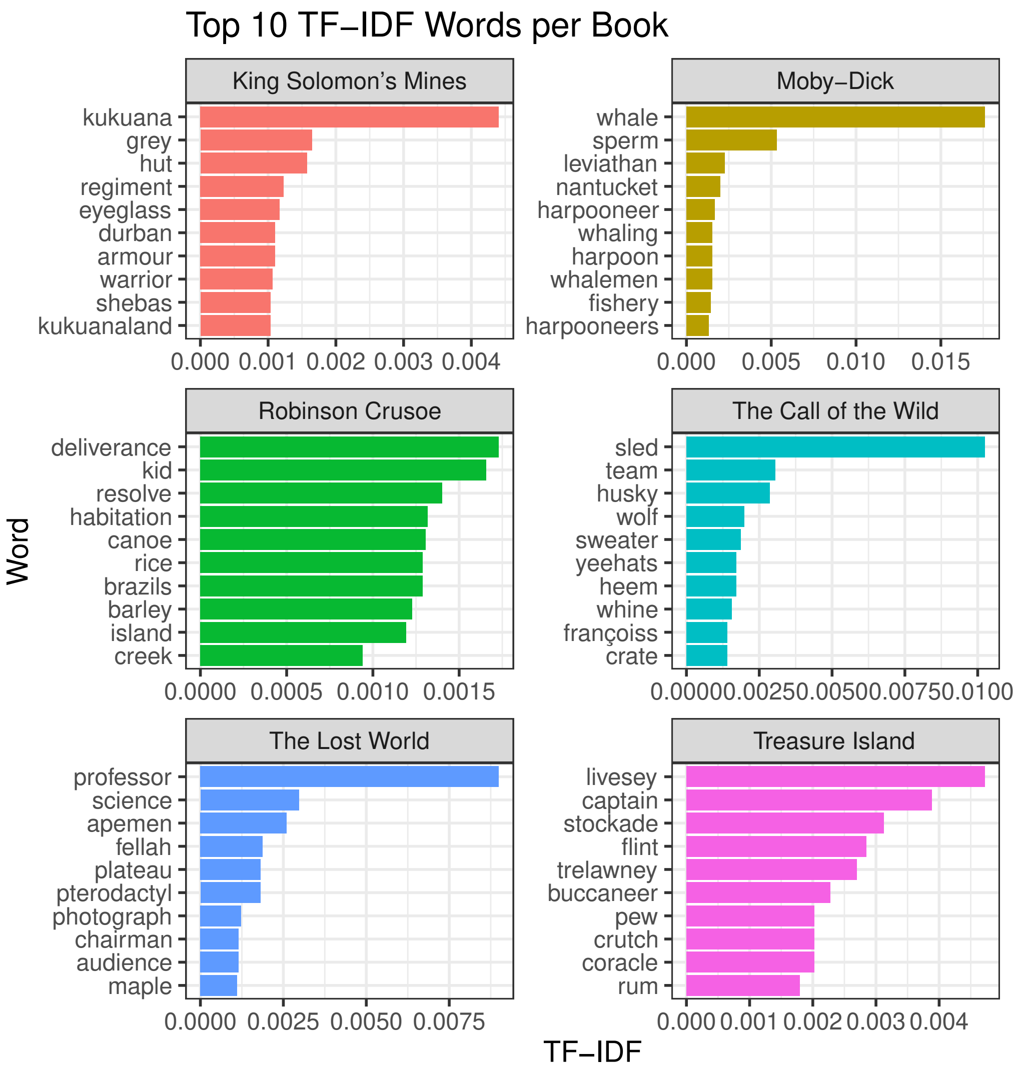

Figure 7: Top 10 words with the highest TF-IDF scores for each of the six novels

Figure 7 highlights the dominant themes across the novels, each of which presents a unique and distinct narrative focus. Words with the highest TF-IDF scores in *King Solomon's Mines* reflect themes that are central to the plot. "Kukuana" refers to the hidden kingdom. Terms like armour, warrior, and regiment emphasise the themes of conflict and violence. "Shebas" refers to the legendary mines of the Queen of Sheba, which are central to the plot. Figure 7 illustrates the dominant theme of whale hunting in *Moby-Dick*. Harpoon, harpooner, whale and whaling reflect the occupational language and the obsessive pursuit of whale. Terms like "habitation", "canoe", and "island" reflect the central theme of being stranded on a deserted island. Additionally, the graph illustrates the character's self-reliance and their efforts to overcome isolation in order to survive in *Robinson Crusoe*. TF-IDF results for call of the wild include words that strongly associate with cold climates. These terms reflect the harsh environment and themes of wilderness. Additionally, it illustrates the character's journey for survival in the wild. Top TF-IDF terms in *The Lost World* reflect the novel's central theme of scientific discovery. Words like "pterodactyl" and "apemen" highlight the discovery of prehistoric creatures and emphasise the observation of new animals and life forms. Additionally, the presence of terms like "fellah" reflects the imperial setting in which the story takes place. Words with the highest TF-IDF scores in *Treasure Island* highlight the novel's maritime and pirate setting. These distinct terms are symbolic of life at sea and pirate culture. This analysis has provided valuable insights into distinct themes of each novel. These results reinforce the defining characteristics of the adventure genre with themes of danger, survival and challenges.

4.3 Topic modelling

To investigate the thematic structure of the six classic adventure novels, Latent Dirichlet Allocation (LDA) was used (see Section 3.4). Two topics were extracted for each novel, offering insight into dominant narrative elements and recurring motifs. By examining the top words associated with each topic ,we identified both book-specific themes and broader adventure genre conventions. This section outlines the key findings from each novel and reflects on cross-cutting thematic patterns.

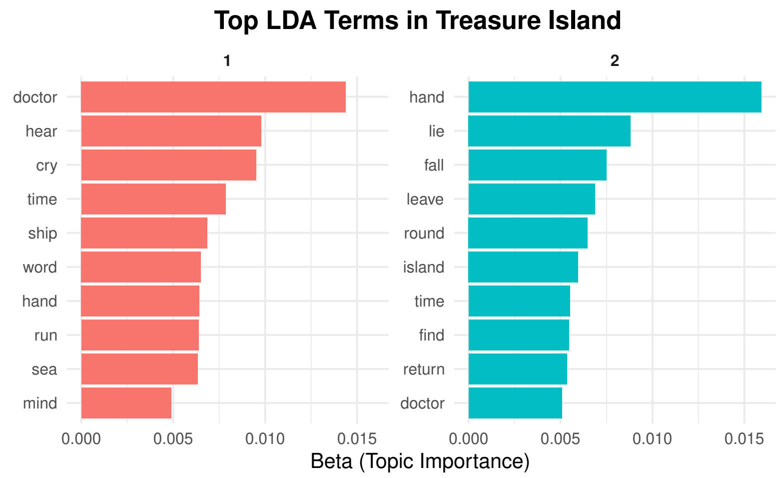

Figure 8: Top 10 LDA terms in *Treasure Island*

From the Figure 8, it is seen that *Treasure Island* is dominated by sailing themes. In Topic 1, words like "doctor," "cry," "run," and "speak" suggest moments of decision, confrontation, and high emotional stakes.Terms like "lie", "fall", "word," and "voice" in Topic 2 point to themes of betrayal, manipulation, and persuasion.

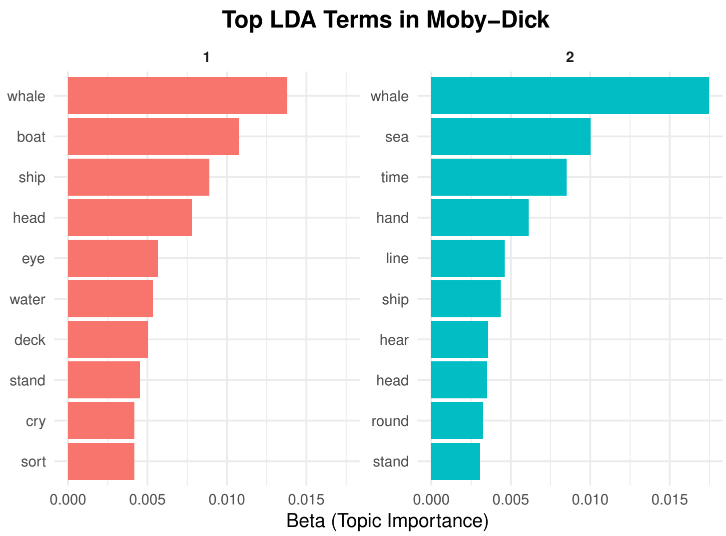

Figure 9: Top 10 LDA terms in *Moby Dick*

As seen in the Figure 9, these topics reflect the strong maritime themes in *Moby-Dick*, with "whale", "ship" and "sea" appearing prominently. Topic 1 emphasises the physical surroundings and the crew's pursuit of the whale. Words like "deck" ,"boat" and "cry" illustrate the chaos that ensues during the chase. Whereas Topic 2 focuses on the actual hunting of the whale. Words like “hand”, “hear”, “round”, ”stand” and “line” imply the hunting process and the crew’s efforts.

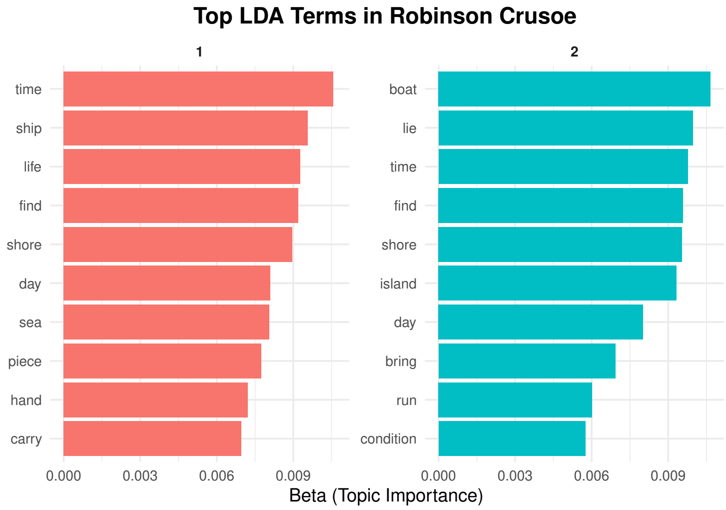

Figure 10: Top 10 LDA terms in *Robinson Crusoe*

From Figure 10, themes of survival, self-reliance, and isolation are prominent in *Robinson Crusoe*. Topic 1 suggests the character’s survival and adaptation to the environment. Words like “carry”, “piece”, “shore” and “find” highlight the daily hardships of surviving on a deserted island. Whereas Topic 2 emphasises the efforts to escape. Terms like “boat,” “island,” and “run” indicate the character’s attempts to escape or leave.

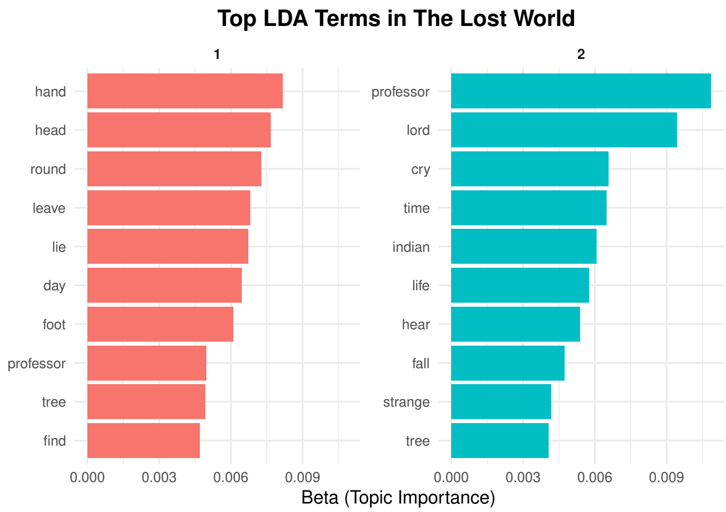

Figure 11: Top 10 LDA terms in *The Lost World*

Themes of exploration and discovery are prominent in *The Lost World* (Figure 11). Topic 1 illustrates the expedition's physical journey. Words like "tree", "find", "day" and "foot" indicate the long journey endured by the characters. Whereas Topic 2 depicts the social roles and emotional themes. Words like "fall", "cry" and "strange" indicate fear and caution among the characters, whereas words like "Indian" and "lord" reflect the colonial setting of the novel.

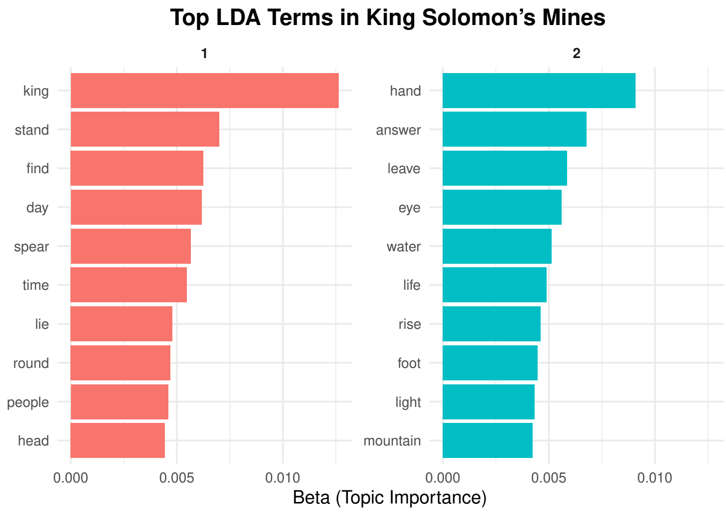

Figure 12: Top 10 LDA terms in King Solomon's Mines

Quest and ancient mystery are the most prominent themes in *King Solomon's Mines* (Figure 12). Topic 1 blends royalty, ritual, and battle. Words like "king", "spear" and "stand" point to the tribal or monarchial society. Whereas Topic 2 illustrates the geographical features and characters' emotions.

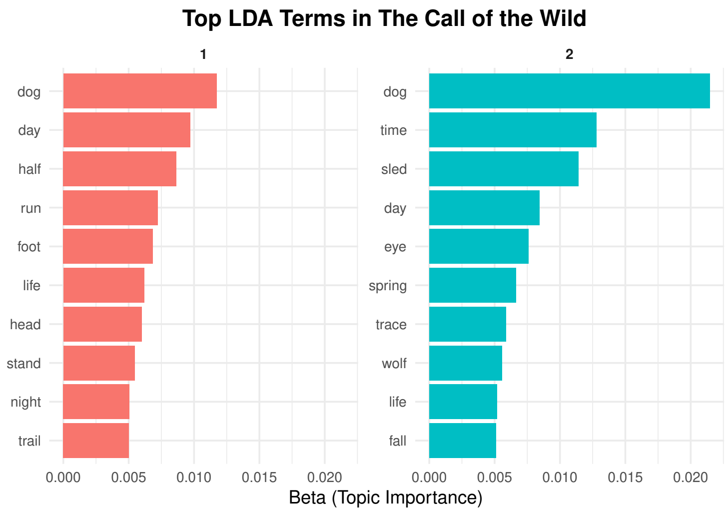

Figure 13: Top 10 LDA terms in *The Call Of The Wild*

Struggle and survival are the most prominent themes in *The call of the wild*(Figure 13) Topic 1 reflects the physical endurance and daily struggle of the protagonist. Terms like "run", "trail" ,"foot" and "night" suggest the constant motion and hardship faced by the characters in the wild. Whereas Topic 2 showcases the protagonist's tragic journey in the wild. Words like "dog", "time", "sled", "day", "spring", "trace" and "wolf" emphasise the struggles of being of adjusting to new surroundings .

The topic modelling results reveal that each novel uses distinct vocabulary to build settings and plots. However, they showcase similar themes including adventure, challenges and survival. The LDA analysis highlights the latent themes in each novel and emphasises the common themes that define the adventure genre.

4.4 Sentiment Analysis

Sentiment analysis offers an overview of the emotional tone throughout the novels. This helps us understand how varying sentiments contribute to shaping the narratives in adventure genre (see Section 3.5).

Table 2 highlights the significant emotional contrasts between the novels. *The Call of the Wild* has the lowest average sentiment score. This may be because of the protagonist's harsh journey of being sold to dog traders, his companions being killed, and being forced to adapt to his harsh surroundings. This novel has very dark themes of death, despair,

Table 2: Overall sentiment score by book

Book	Sentiment score
Robinson Crusoe	0.0666
Moby-Dick	0.0572
Treasure Island	0.0498
King Solomon's Mines	0.0605
The Call of the Wild	0.0407
The Lost World	0.0485

violence, starvation, and abuse which contribute to the low sentiment score.

Robinson Crusoe has the highest sentiment score among these novels This maybe because of the hopeful nature of the story. This graph indicates that the overall sentiment scores for these novels are positive but relatively low. This may be due to the themes of survival, challenges and harsh journeys that the characters undertake in adventure fiction.

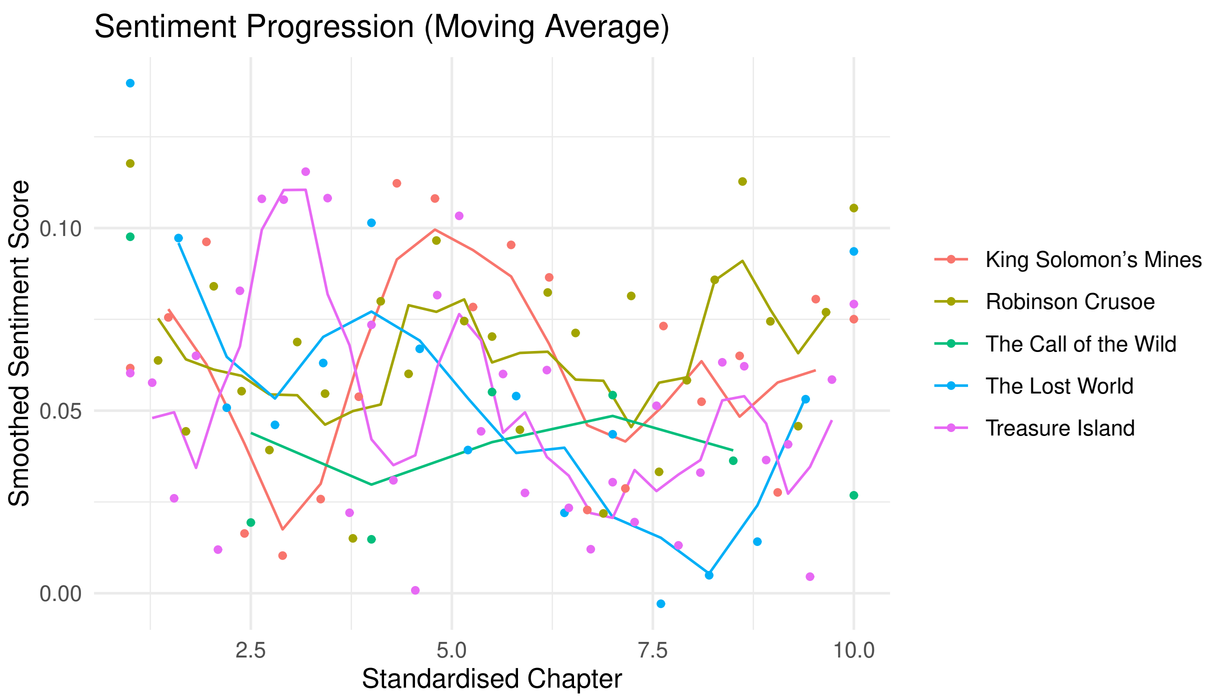

Figure 14: sentiment score progression by standardised chapter across novels

This graph shows how sentiment changes across the standardised chapters of five novels. The sentiment scores are smoothed using a moving average to better show trends over time. Each line represents one novel.

Robinson Crusoe shows the most stable and positive sentiment. The curve remains above the others for most of the book. This matches the hopeful and forward-looking tone of the story, where the character steadily adapts to life on the island.

The Call of the Wild has the lowest scores throughout. The line drops early and stays low until the end. This fits the novel's focus on violence, hardship, and emotional struggle. The low values show a consistently dark tone.

The Lost World starts slightly high but drops sharply in the second half. This suggests a shift in tone as the story moves from excitement and curiosity to danger and fear during the expedition.

Treasure Island shows more variation across chapters. The sentiment rises and falls throughout, reflecting the mix of excitement and betrayal in the plot. There are moments of adventure but also darker turns.

King Solomon's Mines shows an early dip but then rises steadily. This upward trend may reflect growing trust among characters or success in the expedition. However, its curve stays in the middle range compared to the others.

Overall, the sentiment curves show how each novel's emotional tone changes over time. *Robinson Crusoe*stays the most positive. *The Call of the Wild* is the most negative. The other novels show a mix of rises and falls, reflecting shifts in action, tone, and emotion as the stories progress.

All novels show some degree of change after the first 25^{th} percentile of the book. *The Lost World*, *Robinson Crusoe*, and *King Solomon's Mines* showcase a dip in the sentiment after 25^{th} percentile of book, whereas, *Moby-Dick* and *Treasure Island* show increases sentiment after the first 25^{th} percentile. Similarly, the sentiment scores change noticeably around the 50^{th} percentile (i.e. halfway through the book). *Treasure Island*, *Robinson Crusoe* and *King Solomon's Mines* have a peak around this midpoint and then gradually decrease until the final quartile begins.

In the last quartile all novels except for *The Call of the Wild* end on a relatively high note.

This pattern suggests the structure of adventure novels, where the protagonist's journey is shaped by a series of challenges. This sentiment progression mirrors the storytelling structure of the adventure genre, where the characters undertake a tumultuous journey and overcome struggles as the narrative unfolds.

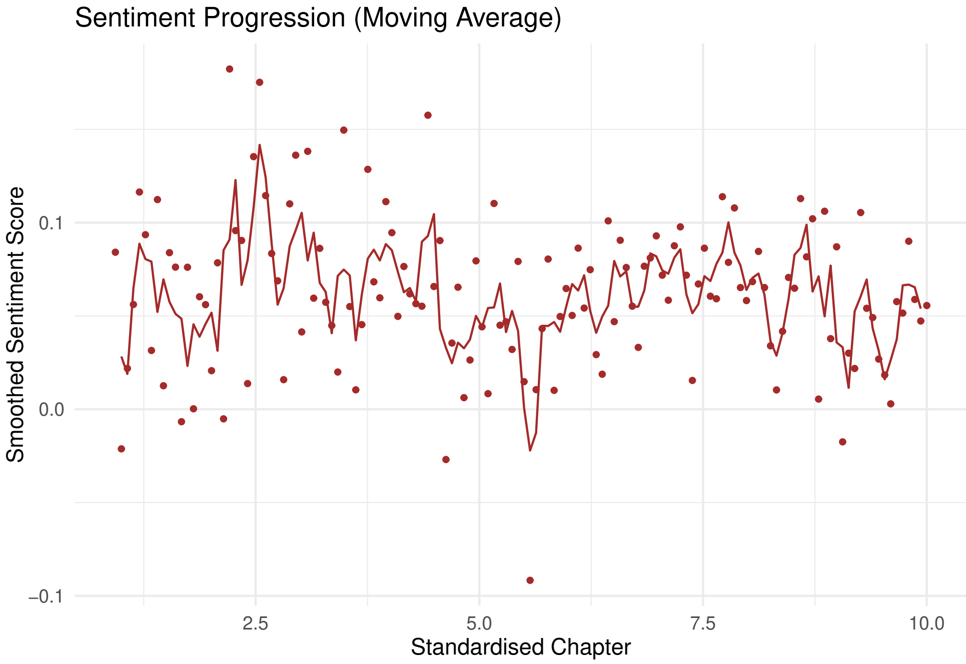

Figure 15: *Moby-Dick* sentiment progression

Moby-Dick is plotted separately because of its unique structure. This graph fluctuates a lot due to over 130 chapters being condensed into just 10. However, it follows similar characteristics, with the book reaching a high around the 25 percentile and it reaches its lowest point after 50$^{\text{th}}$ percentile. Despite its unique structure, *Moby-Dick*showcases the characteristics of adventure genre and confirms to the narrative structure of the adventure genre.

4.5 NRC Word Emotions

The NRC lexicon was used to calculate the proportion of of each emotion for the six novels (see Section 3.6)

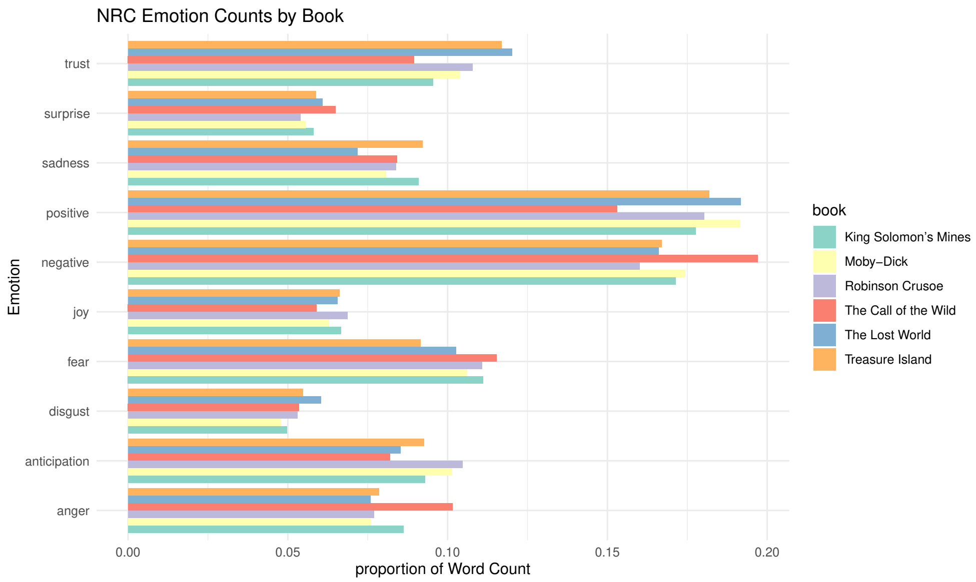

Figure 16: Proportion of emotions across books

From (Figure 16) we note that Surprise was low across all novels. This suggests that while the stories have unexpected events, they are not described using strong language related to surprise. The plots are more focused on steady progress and conflict than on sudden twists. Anticipation was high in *Robinson Crusoe*. This matches the story's focus on planning, building, and surviving on the island. In contrast, anticipation was low in *The Call of the Wild*, where the story focuses more on reaction than planning. *The Call of the Wild* also had the highest negative sentiment and strong anger scores. This supports the idea that the book has very dark themes. These include violence, death, fear, and suffering. The language in the book reflects the harsh experiences of the main character. Trust was high in *Treasure Island* and *The Lost World*. This fits with the presence of group journeys and cooperation. In Treasure Island, characters like Jim and the doctor work together. In *The Lost World,* trust is important among the explorers. Joy was low in all six novels. This shows that although these books are about adventure and discovery, they are not joyful stories. Most of the books focus on danger, struggle, and survival.

Low joy and relatively high anticipation and fear in these novels conform to the characteristics of the adventure genre, where protagonists must show a degree of caution and face journeys full of danger and challenges.Sadness being relatively higher than joy indicates the difficulties the characters experience. This showcases the challenges that the characters face, such as the death of friends or crew members, loss and other hardships. Additionally, varying levels of trust indicate how different storylines include caution in their journeys. These tropes reflect the defining characteristics of the adventure genre.

4.6 Protagonist Mentions

Analysing the frequency of protagonist mentions provides insights into the narrative structure. The proportion of protagonist mentions was calculated for each of the six novels (see Section 3.7).

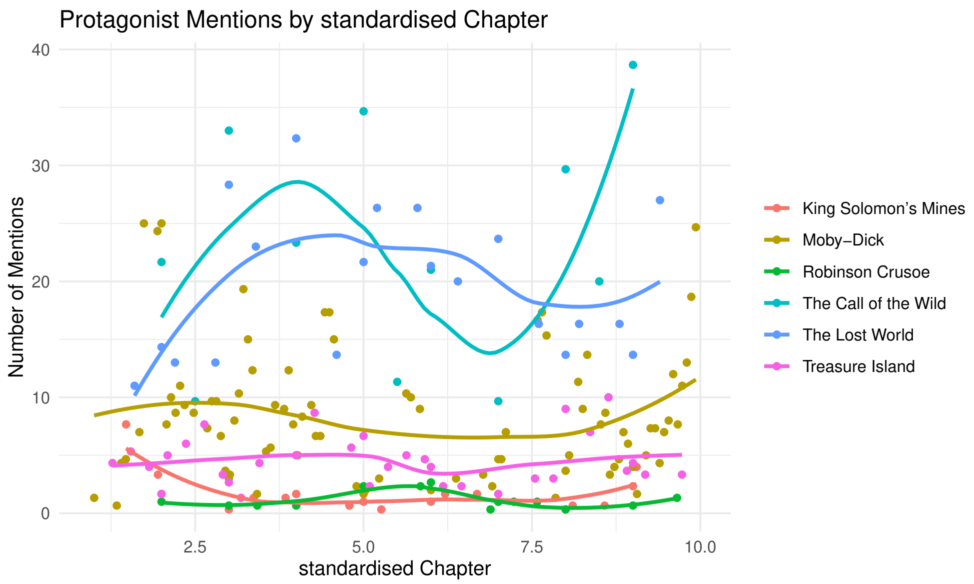

Figure 17: Protagonist mentions by standardised chapter across novels

Figure 17 displays the moving average plot of protagonist mentions across the six novels. This graph reveals clear differences in how the main characters are referenced throughout the narratives. *The Call of the Wild* shows the highest overall level of mentions, with Buck's name appearing consistently across almost all chapters. This pattern reflects a narrative structure in which the protagonist remains at the centre of the action from beginning to end, aligning with the adventure genre's tendency to focus closely on the hero's journey.

The Lost World follows a similar pattern, with frequent mentions of its main characters. The consistency of these mentions highlights how their presence drives the story forward.

After these two, there is a noticeable drop before *Moby-Dick* and *Treasure Island*. Both of these novels show more fluctuation in protagonist mentions across chapters.

King Solomon's Mines , *Treasure Island* and *Robinson Crusoe* have the lowest protagonist mentions overall. In *King Solomon's Mines*, this may be due to the nature of the story, where attention is shared between a group of characters. Additionally, in these novels the protagonist is the narrator, hence it may reduce the explicit mentions despite the narrative being entirely centred on his survival and exploration.

This analysis shows that novels where the main character is also the narrator have significantly fewer mentions. This may be because this analysis only considers character names and aliases, not pronouns. Overall, these variations show the importance of the protagonist to the storyline in adventure fiction.

5 Conclusion

This report provided an analysis of the thematic elements of the selected adventure novels. NLP techniques like word frequencies, word clouds, TF-IDF, topic modelling, and sentiment analysis were applied to these books. By applying these methods, we were able to uncover shared themes and distinct features that characterize the adventure genre.

The word clouds identified common themes of exploration, adventure, and survival. TF-IDF and topic modelling helped uncover themes unique to each novel. Furthermore, sentiment analysis emphasised these themes.

These novels all shared a common theme of adventure and being on a journey, but each novel explored it differently. *Treasure Island* and *Moby-Dick* focus on a maritime journey and a pursuit of goals. In contrast, *The Call of the Wild* and *Robinson Crusoe* share common themes of survival, isolation, and self-reliance. *King Solomon's Mines* and *The Lost World*, on the other hand, share themes of discovery and exploration along with caution.

The sentiment analysis and word cloud confirmed the dark themes of violence, abuse, death, and tragedy in *The Call of the Wild*. Additionally, TF-IDF and word clouds showcased *Moby-Dick*'s theme of obsession with the hunt and the whale. Furthermore, themes of caution and fear were common in *The Call of the Wild*, *King Solomon's Mines*, and *The Lost World*.

These novels share similar themes and narrative structures, despite being written across different time periods. This suggests that the core elements of the adventure genre have remained consistent over time, even as each author explores them in a unique way. These methods could provide better insights into the evolution of adventure genre.

Together, these techniques provided a comprehensive view of how classic adventure novels explore themes of exploration, discovery, survival and pursuit. Viewed across decades, these findings suggest that the central identity of the adventure genre remains the same. Each author brings their own focus, whether it is danger or the thrill of discovery. These results highlight how text mining methods can help uncover patterns in language that reflect deeper themes in literature.

A Appendices

A.1 List of terms removed from the dataset

- jim
- hawkins
- silver
- squire
- tom
- joyce
- abraham
- alan
- hispaniola
- ben
- gunn
- morgan
- smollett
- im
- ive
- long
- john
- captain
- redruth
- bildad
- loo
- thy
- xury
- friday
- incubu
- bougwan
- ventvögel
- gladys
- zambo
- waldron
- malone
- mcardle
- challenger
- summerlee
- roxton
- white
- quatermain
- umbopa
- twala
- gagool
- scragga
- infadoos
- curtis
- kafir
- foulata
- imotu
- khivs
- sir
- henry
- ignosi
- kraal
- twalas
- zulu
- macumazahn
- solomons
- you
- aye
- henrys
- solleks
- miller
- manuel
- françois
- perrault
- thornton
- mercedes
- thorntons
- thornton's
- hans
- charles

- billee
- dawson
- pete
- hal
- buck
- curly
- spitz
- dave
- pike
- dub
- leks
- ha
- ishmael
- ahab
- moby
- stubb
- flask
- ahabs
- thee
- thou
- ye
- solomon
- queequeg
- starbuck
- pequod
- pip
- peleg

A.2 Word normalisation and replacement list

Table 3: Word Normalisation and Replacement

Original Words	**Replaced With**
dead, die	death
cap'n	captain
whale's	whale
live, living	life
kukuana's, kukuanas	kukuana
scientific	science

A.3 Protagonist mentions by chapters

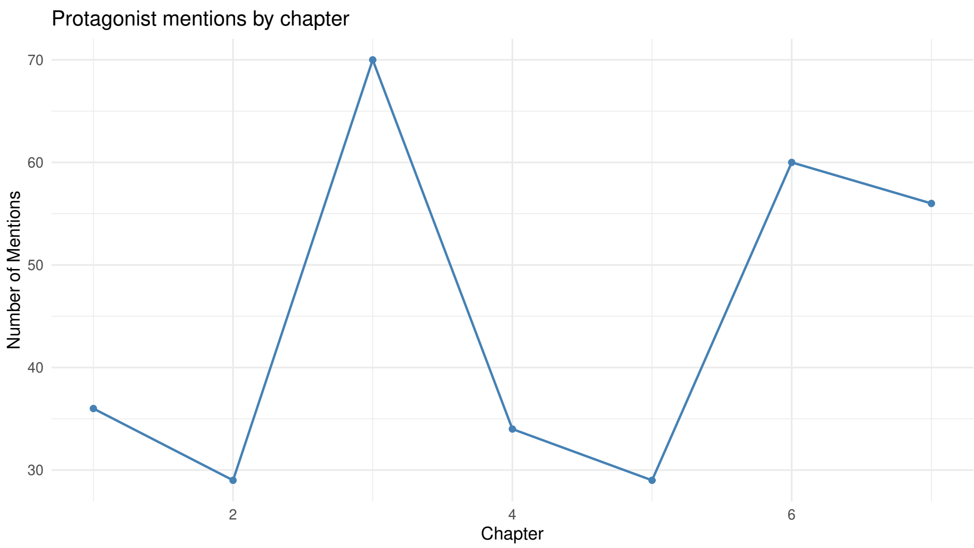

Figure 18: *The Call of the Wild*

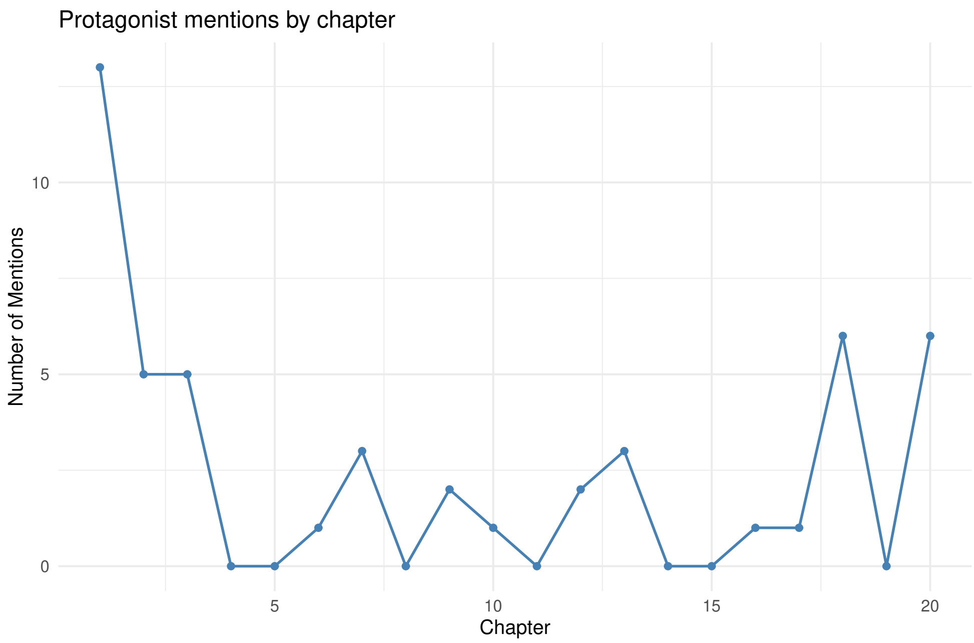

Figure 19: *King Solomon's Mines*

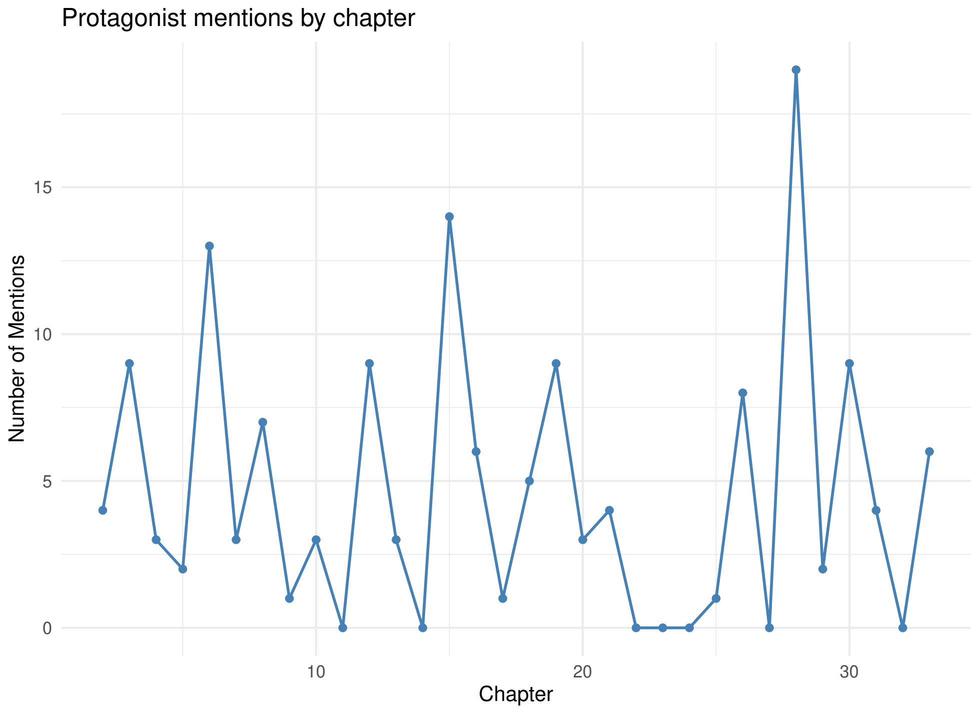

Figure 20: *Treasure Island*

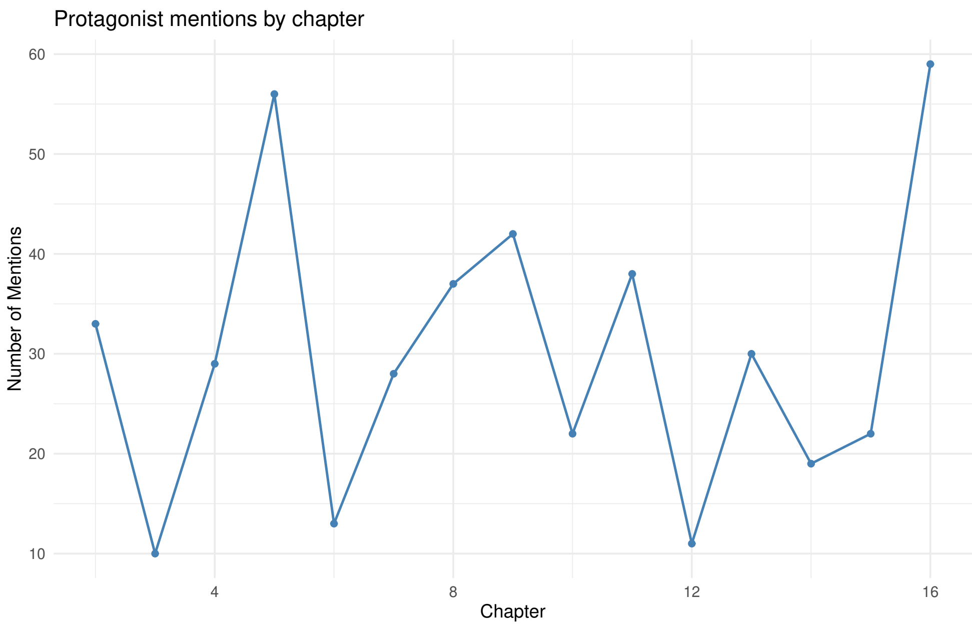

Figure 21: *The Lost World*

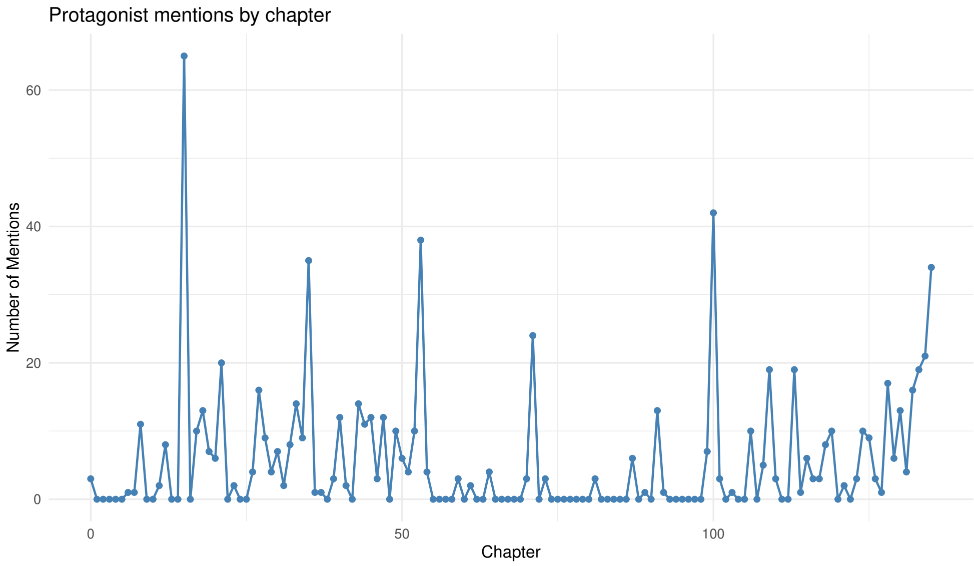

Figure 22: *Moby-Dick - ahab*

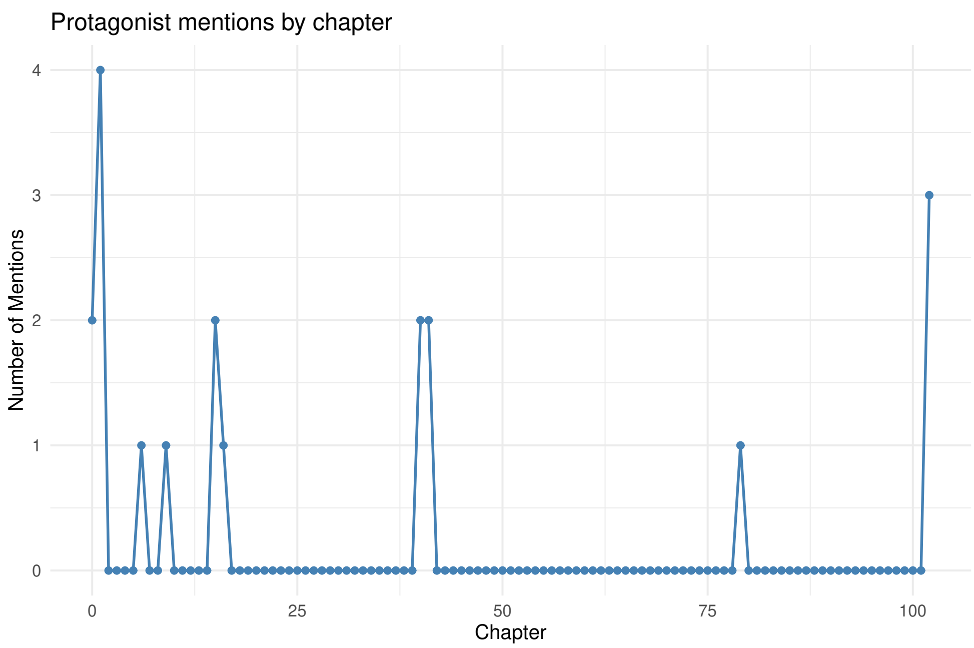

Figure 23: *Moby-Dick - Ishmael*

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